9.8 Seed Protection

Seed is one of the most crucial elements in the livelihoods of agricultural communities. It is the repository of the genetic information of crop species and their varieties resulting from the continuous improvement and selection over the time. The potential benefits of seed to crop productivity and food security can be enormous. Crop protection products can be applied during the growth of the crop or added to the seed. Seed protection play a significant role in improving the establishment of healthy crops. Protection and storage of seeds can be done by traditional and modern methods.

9.8.1. Traditional methods of Seed Protection

- In traditional method seeds are coated with fine red soil, Guntur Chilli Powder, Neem leaf Powder, Powder of Bittergourd, Drumstick extract, Pongamia leaf extract and stored for short duration.
- Paddy Seeds are immersed in salt water in the ratio of 1:10 to remove the floating chaffy seed and then dried in shade for one -two years of storage.
- Sorghum seeds were treated with lime water (1 kg of lime in 10 litres of H_20) for 10 days and then the seeds are dried and stored.
- Chickpea were treated with citronella leaf oil, cotton seed oil, soya bean oil, castor seed oil (500 ml of oil for 100 kg of seed).
- Sunflower seeds were kept inside the dried fruit of sponge gourd after removing the seeds. These fruits were kept in an airtight container.

9.8.2. Modern Methods of Seed Protection

The various modern methods of seed protection are as follows:

1. Seed Treatment

In agriculture and horticulture, seed treatment or seed dressing is a chemical, typically antimicrobial or antifungal, with which seeds are treated (or dressed) prior to planting.

Benefits of seed treatment

- Prevents spread of plant disease.
- Protects seed from seedling blights.
- Improves germination.
- Provides protection from storage insects.
- Controls soil insects.

2. Seed Hardening

Seed hardening is a physiological preconditioning of the seed by soaking of seed in water or chemical solution for definite duration in proper ratio (Seed : Solution) and shade drying to bring back the seed to original moisture content.

Benefits:

- It increases the yield, root growth and vigour of seed germination
- The uniformity of seedling emergence.
- Flowering occurs 2-3 days earlier
- Uniform seed set and maturity
- Exposes the seed to drought tolerance.

3. Seed Pelleting

The process of enclosing seed in a filter inert material using an adhesive with bioactive chemicals. Seed pelleting increases the weight, size and shape of seeds by allowing percale maturing and spacing of seed in the field.

4. Seed coating

Seed coating is a thicker form of covering of seed and may contain fertilizer, growth promoters, rhizobium inoculum, nutritional elements and repulsive agents. Chemical, pesticides added to the seed by adhesive agents cause increased seed performance and seed germination.

5. Bio Priming of Seeds

Bio-Priming is a process of biological seed treatment that refers to combination of seed hydration (physiological aspect of disease control) and inoculation (biological aspect of disease control) of seed with beneficial organism to protect seed. It is an ecological approach using selected fungal antagonists against the soil and seed borne pathogens. This seed treatments may provide an alternative to chemical control.

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